HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained **Linux kernel** modules and device drivers for embedded systems, focusing on stability and performance improvements across various hardware architectures.
- Debugged complex **kernel-level issues** including panics, hangs, and performance regressions using tools such as **ftrace**, **crash**, and **gdb** on RHEL-based environments.
- Engineered and operated **CI/CD pipelines** for kernel development and validation, automating build, test, and release processes for multiple Linux distributions.
- Collaborated with cross-functional teams including SRE, platform, and hardware engineers to shepherd kernel changes into production and qualify new hardware at the kernel layer.
- Contributed to upstreaming device drivers and implementing kernel tuning for stability and performance across core OS subsystems like memory management and scheduling.
- Authored **Python scripts** for automating data validation and reporting tasks related to kernel release quality assurance, reducing manual effort by an estimated **30%**.
- Managed **Docker container** deployments and basic Kubernetes configurations for development environments, ensuring consistent build and test execution.

SKILLS

Kernel & System Software – Linux Kernel, Kernel Modules, Device Drivers, System Software, Kernel Debugging, Kernel Tuning, RHEL, RPM Packaging, Core OS Components

Programming Languages – C, Python, Bash, Rust, Go

Debugging & Profiling Tools – perf, ftrace, bpftrace, crash, gdb

CI/CD & Automation – CI/CD Systems, Jenkins, Git, Shell Scripting

Operating System Concepts – Processes, Memory Management, Filesystems, Networking, Scheduling, Interrupts, Hardware Interactions

Containerization & Orchestration – Docker, Kubernetes

PROJECTS

Linux Kernel CI and Validation Framework

- Designed and implemented an automated CI system using **Jenkins** and **Bash scripting** to compile, test, and validate Linux kernel builds across different hardware targets.
- Integrated **ftrace** and **perf** for performance analysis and regression detection, identifying and resolving latency spikes in critical kernel subsystems.
- Developed reporting mechanisms to track build success rates and identified regressions, providing actionable insights to the kernel development team.

Embedded Linux Driver Development & Debugging

- Developed and debugged custom **device drivers** in **C** for a proprietary hardware interface within an embedded Linux environment.
- Utilized **gdb** and kernel debugging techniques to resolve race conditions and memory corruption issues, improving system stability.
- Documented driver architecture and debugging procedures, contributing to the team's knowledge base for future development.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)